FITS Image Combiner Tutorial: Image Stacking and Calibration

Overview

The FITS Image Combiner (`fits-image-combiner.py`) is a sophisticated tool for combining multiple astronomical images to improve signal-to-noise ratio (SNR), remove cosmic rays, and create high-quality calibration frames. It automatically aligns images using star detection and offers multiple statistical combination methods to handle different types of noise and artifacts.

Why Combine Images?

Signal-to-Noise Improvement

When you combine N images, the signal adds linearly while random noise adds in quadrature:

- **Signal increases by factor of N**
- **Noise increases by factor of √N**
- **SNR improves by factor of √N**

Example: Combining 16 images improves SNR by 4×, making faint objects visible that were buried in noise in individual frames.

Cosmic Ray Rejection

Cosmic rays create bright spots in random locations. By combining multiple images with outlier rejection, these artifacts are automatically removed while preserving real astronomical signals.

Creating Master Calibration Frames

Multiple flat fields or dark frames are combined to create low-noise master calibration frames that don't add noise to your science images.

Installation Requirements

```
pip install numpy scipy astropy photutils
```

Combination Methods

The program offers four statistical methods for combining images:

1. Average (`-c average`)

How it works: Simple mean of all pixel values

```
combined_pixel = (image1 + image2 + ... + imageN) / N
```

Best for:

- High SNR images without outliers
- Creating master bias frames
- Quick combinations when cosmic rays are minimal

Pros: Maximum SNR improvement (√N) **Cons**: Cosmic rays and hot pixels create bright spots

2. Median (`-c median`)

How it works: Takes the middle value when pixels are sorted

```
pixel_values = [102, 105, 108, 195, 110]  # 195 is cosmic ray
sorted = [102, 105, 108, 110, 195]
median = 108  # cosmic ray rejected!
```

Best for:

- Images with many cosmic rays
- Creating robust master darks
- When you have 5+ images

Pros: Excellent cosmic ray rejection **Cons**: Lower SNR improvement (~0.8×√N), needs odd number of images for best results

3. Sigma-Clipped Average (`-c avsigclip`)

How it works: Removes pixels more than N-sigma from median, then averages remaining

```
# With -N 3.0 (3-sigma clipping)
pixel_values = [100, 102, 105, 108, 250]  # 250 is outlier
median = 105
sigma = 3.8
# 250 is >3-sigma away, rejected
average = (100 + 102 + 105 + 108) / 4 = 103.75
```

Best for:

- Master flat fields (with `-N 2.5`)
- Science images with moderate cosmic rays
- Optimal balance of SNR and outlier rejection

Pros: Good SNR while rejecting outliers **Cons**: Requires tuning sigma parameter

4. No-Maximum (`-c nomax`)

How it works: Removes the brightest pixel value at each position, averages the rest

```
pixel_stack = [100, 102, 105, 280]  # 280 could be cosmic ray
# Remove maximum (280)
average = (100 + 102 + 105) / 3 = 102.33
```

Best for:

- Quick cosmic ray removal with 3-4 images
- Creating flats from limited data
- When cosmic rays always brighten pixels

Pros: Simple, effective for bright outliers **Cons**: Always loses one data point per pixel

Image Alignment

Automatic Star Matching (Default)

The program automatically detects and matches stars between images to correct for telescope drift:

```
# Process science images with automatic alignment
python3 fits-image-combiner.py NGC7331/ -o ngc7331_combined.fits
```

How it works:

1. Detects up to 20 brightest stars using DAOStarFinder
2. Matches star patterns between reference and target images
3. Computes average shift (dx, dy)
4. Applies sub-pixel shifts using cubic interpolation

Output shows the alignment process:

```
Finding stars in reference image:
  Star 1: x=1205.32, y=892.15
  Star 2: x=2341.87, y=1456.23
  ...

Aligning file: NGC7331_002.fits
  Matched Star 1: Ref(1205.32,892.15) Target(1207.45,894.28) Shift(dx=-2.13, dy=-2.13)
  Applying shift: dy=-2.13, dx=-2.13
```

Disabling Alignment (`--noalign`)

Use for calibration frames where stars shouldn't be aligned:

```
# Combine flat fields without alignment
python3 fits-image-combiner.py r_flats/ -c avsigclip --noalign --normalize
```

When to use `--noalign`:

- **Flat fields**: Uniform illumination, no stars to align
- **Dark frames**: No light, only thermal signal
- **Bias frames**: No signal at all
- **Sky flats**: Gradient changes between frames

Creating Master Calibration Frames

Master Flat Field

Flat fields correct for pixel sensitivity variations and vignetting:

```
# Optimal flat field combination
python3 fits-image-combiner.py r_flats/ \
    -c avsigclip \
    -N 2.5 \
    --noalign \
    --normalize \
    -o master_r_flat.fits
```

Parameters explained:

- `-c avsigclip`: Removes dust specks and outliers
- `-N 2.5`: 2.5-sigma clipping (tighter than default 3.0)
- `--noalign`: Flats have no stars to align
- `--normalize`: Scales output to median=1.0 for proper calibration
- `-o master_r_flat.fits`: Descriptive output name

Alternative using no-max method:

```
# When you have fewer flats (3-5 images)
python3 fits-image-combiner.py r_flats/ \
    -c nomax \
    --noalign \
    --normalize \
    -o rflat.new.fits
```

Master Dark Frame

Dark frames capture thermal noise and hot pixels:

```
# Combine dark frames
python3 fits-image-combiner.py darks/ \
    -c median \
    --noalign \
```

```
    -o master_dark_300s.fits
```

- Use `median` for robust hot pixel rejection
- No normalization needed for darks
- Name should include exposure time

Master Bias Frame

Bias frames capture readout noise:

```
# Combine bias frames (very low noise, use average)
python3 fits-image-combiner.py bias/ \
    -c average \
    --noalign \
    -o master_bias.fits
```

Processing Science Images

Example: NGC 7331 Galaxy Stack

Complete workflow for combining galaxy images:

Step 1: Create Master Calibration Frames

```
# Master bias
python3 fits-image-combiner.py bias/ -c average --noalign -o master_bias.fits

# Master dark (300s exposure)
python3 fits-image-combiner.py darks_300s/ -c median --noalign -o
master_dark_300s.fits

# Master flat for r-band
python3 fits-image-combiner.py r_flats/ -c avsigclip -N 2.5 --noalign --
normalize -o master_r_flat.fits
```

Step 2: Combine Science Images with Calibrations

```
# Combine NGC 7331 images with full calibration
python3 fits-image-combiner.py NGC7331_r_band/ \
    -dark master_dark_300s.fits \
    -flat master_r_flat.fits \
    -c avsigclip \
    -N 3.0 \
    -o NGC7331_r_combined.fits
```

What happens:

1. Each image is dark-subtracted: `calibrated = raw - dark`
2. Each image is flat-fielded: `calibrated = calibrated / flat`
3. Stars are detected and images aligned to first frame

4. Aligned images are sigma-clipped and averaged
5. Final stacked image saved with improved SNR

Example Output

```
Aligning file: NGC7331_002.fits
Finding stars in reference image:
  Detected 20 stars. Centroids (x, y):
  Star 1: x=2453.21, y=1832.44
  Star 2: x=1205.67, y=2341.18
  ...

Finding stars in target image:
  Detected 18 stars. Centroids (x, y):
  Star 1: x=2455.33, y=1834.52
  ...

  Matched Star 1: Shift(dx=-2.12, dy=-2.08)
  Matched Star 2: Shift(dx=-2.09, dy=-2.11)
  Applying shift: dy=-2.10, dx=-2.10

Combining 15 images using avsigclip method...
Final combined image saved as 'NGC7331_r_combined.fits'

Output image statistics:
  Min: 95.2341
  Max: 18934.5234
  Mean: 512.3421
  Median: 487.2134
```

Understanding Normalization

The `--normalize` flag is crucial for creating proper flat fields:

Without Normalization

```
python3 fits-image-combiner.py flats/ -c median --noalign
# Output median might be 20,000 ADU
```

When you divide science images by this flat, values become very small.

With Normalization

```
python3 fits-image-combiner.py flats/ -c median --noalign --normalize
# Output median = 1.0
```

Science images maintain proper scaling after flat field correction.

How normalization works:

- For `-c average`: Divides by mean of all pixels
- For `-c median`: Divides by median of all pixels
- For `-c avsigclip`: Divides by mean after clipping
- For `-c nomax`: Divides by mean after removing maxima

Advanced Examples

Combining Images from Multiple Nights

```
# First align all images to common reference
python3 fits-image-combiner.py all_nights/ \
    -flat master_flat.fits \
    -c avsigclip \
    -N 2.0 \
    -o combined_all_nights.fits
```

Creating a Deep Sky Image

```
# Aggressive cosmic ray rejection for deep imaging
python3 fits-image-combiner.py deep_field/ \
    -dark master_dark.fits \
    -flat master_flat.fits \
    -c median \
    -o deep_field_combined.fits
```

Quick Look Combination

```
# Fast combination for checking data at telescope
python3 fits-image-combiner.py tonight/ \
    -c average \
    -o quick_look.fits
```

Troubleshooting

"No stars found!"

- Image may be out of focus
- Field may contain only faint stars
- Try using `--noalign` if alignment isn't needed

"No matched stars found"

- Images may have large shifts (>15 pixels)
- Field rotation between images
- Different fields accidentally mixed

Alignment introduces black borders

- Normal behavior when images are shifted
- Crop final image to remove borders
- Take more images with better tracking

Output has strange patterns after flat fielding

- Ensure flats are normalized (`--normalize`)
- Check flat was taken with same optical configuration
- Verify flat has good SNR (combine more flats)

Best Practices

1. **Always use calibration frames** when available
2. `-dark master_dark.fits -flat master_flat.fits`
3. **Choose combination method based on data**:
   - Few images (3-5): Use `median` or `nomax`
   - Many images (10+): Use `avsigclip`
   - No cosmic rays: Use `average` for best SNR
4. **Tune sigma clipping**:
   - Flats: `-N 2.5` (tighter clipping)
   - Science: `-N 3.0` (standard)
   - Bad conditions: `-N 2.0` (aggressive)
5. **Organize your data**:
6. `project/`
7. `├── bias/`
8. `├── darks/`
9. `├── flats_r/`
10. `├── flats_g/`
11. `├── science_r/`
12. `└── science_g/`
13. **Name outputs descriptively**:
14. `-o NGC7331_r_band_avsigclip_2024-08-15.fits`

Summary

The FITS Image Combiner transforms multiple noisy images into a single high-quality result through intelligent alignment and statistical combination. Whether creating pristine calibration frames or revealing faint galaxies, understanding each combination method and parameter helps you extract maximum science from your data. The automatic star alignment ensures precise registration while multiple statistical methods handle different noise characteristics, making this tool essential for modern astronomical image processing.